Topology

Problem Set 2

Question No. 1

Consider the set

$X = \mathbb{R}$ equipped with cofinite topology

$\tau = \{\phi\} \cup \{U \subseteq \mathbb{R} : \mathbb{R} \setminus U \text{ is finite}\}$

that is, the open sets are the empty sets or subsets of $\mathbb{R}$ whose complements are finite.

Now, we know that a topological space is T_1 if every singleton set is closed

Let $x \in \mathbb{R}$

then $\mathbb{R} \setminus \{x\}$ is ~~closed~~ open because its complement $\{x\}$ is finite

Therefore, $\{x\}$ is closed

Since x is arbitary, every singleton subset of $\mathbb{R}$ closed

Hence, $(\mathbb{R}, \tau)$ is a T_1 space

Consider the constant sequence

for $x_n = n, n \in \mathbb{N}$

$x_n \to x$ if every of open neighbourhood U of

$\exists N \in \mathbb{N}$ such that

$n \geq N \Rightarrow x_n \in U$

Let $x \in \mathbb{R}$ be arbitary

Let U be any open neighbour hood of x

By the definition of cofinite topology $\mathbb{R} \setminus U$ is finite

Since $x_n = n$, the sequence consists of distinct natural numbers.

Consequently only finitely Many terms of the sequence can belong to the finite set $\mathbb{R} \setminus U$

Therefore, $\exists N \in \mathbb{N}$ such that

$n \geq N \Rightarrow n \notin \mathbb{R} \setminus U$

thus, $n \geq N \Rightarrow x_n = n \in U$

So, By the definition of convergence

$x_n \to x \quad \forall x \in \mathbb{R}$

Now, the sequence $x_n = n$ converges to every real number

$\Rightarrow$ its limit is not unique

$\Rightarrow$ Space cannot be Hausdroff

thus $(\mathbb{R}, \tau)$ is T_1 but not T_2

$\Rightarrow T_1 \not\Rightarrow T_2$

Question No. 2

Let $U = U_1 \times U_2 \times \cdots \times U_n$ be a basic open set in product topology

take an arbitary point

$$x = (x_1, x_2, x_3, x_4 \cdots x_n) \in U$$

Since each U_i is open in $\mathbb{R}$

$$\exists\, \varepsilon_i > 0 \text{ such that}$$

$$(x_i - \varepsilon_i, x_i + \varepsilon_i) \subseteq U_i$$

choose $\varepsilon = \min\{\varepsilon_1, \varepsilon_2 \cdots \varepsilon_n\}$

Since there are finitely many positive numbers

$$\varepsilon > 0$$

Now, Consider any point

$$y = (y_1, y_2 \cdots y_n) \in B(x, \varepsilon)$$

then $d(x, y) < \varepsilon$

thus, $$\sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} < \varepsilon$$

$$\Rightarrow \sum_{i=1}^{n} (x_i - y_i)^2 < \varepsilon^2$$

Since everyterm in the sum is non-negative

$$(x_i - y_i)^2 < \varepsilon^2$$

for every $i = 1, 2, \dots n$

Consequently $|x_i - y_i| < \varepsilon \leq \varepsilon_i$

thus, $y_i \in (x_i - \varepsilon_i, x_i + \varepsilon_i) \subseteq U_i$

this holds for every i. Hence

$$y \in U_1 \times U_2 \dots U_n$$

$$\Rightarrow B(x, \varepsilon) \subseteq U$$

Since $x \in U$ was arbitary, every point of U has a Euclidean open ball contained in U.

Hence U is Euclidean open.

$\tau_{pro} \subseteq \tau_{Euc}$

Let V be an arbitary Euclidean open subset of $\mathbb{R}^n$

take any point

$$x = (x_1, x_2 \dots x_n) \in V$$

Since V is Euclidean open, $\exists\, \gamma > 0$ such that

$$B(x, \gamma) \subseteq V$$

Now consider the product of open intervals

$$W = \prod_{i=1}^{n} \left(x_i - \frac{\gamma}{\sqrt{n}}, x_i + \frac{\gamma}{\sqrt{n}} \right)$$

We will show that

$$W \subseteq B(x, r)$$

take any $y = (y_1, y_2, y_3, y_4, \ldots y_n) \in W$

By definition of W

$$|x_i - y_i| < \gamma/\sqrt{n} \quad \forall\ i = 1, 2, 3, 4 \ldots, n$$

therefore

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

$$< \sqrt{\sum_{i=1}^{n} \frac{\gamma^2}{n}}$$

$$= \sqrt{\frac{n\gamma^2}{n}}$$

$$= \gamma$$

thus $y \in B(x, \gamma)$

Consequently $W \subseteq B(x, \gamma) \subseteq V$

Notice that W is a basic open set in the product topology because it is a product of open interval in $\mathbb{R}$.

Hence, every point $x \in V$ has a basic product open neighbourhood contained in V.

therefore, V is open in product topology

so, $\tau_{Euc} \subseteq \tau_{prod}$

So $\tau_{prod} = \tau_{EUd}$

Question No. 3

We define $f: X \to X\phi$

by $f((x_\alpha)_{\alpha \in A}) = (x_{\phi(\alpha)})_{\alpha \in A}$

Since ϕ is a bijection, every coordinate in the second product corresponds to exactly one coordinate in the first product.

Since ϕ is bijective, it has an inverse

$\phi^{-1}: B \to A$

define $g: X_0 \to X$

by $g((y_\beta)_{\beta \in B}) = (y_{\phi^{-1}(\beta)})_{\beta \in B}$

then for every $x = (x_\alpha) \in X$

$$g(f(x)) = x$$

Similarly for every $y \in X_0$

$$f(g(y)) = y$$

thus $g = f^{-1}$

Hence f is bijective.

the product topology has a sub basis consisting of

sets of the form

$\pi_\alpha^{-1}(U_\alpha)$

Where U_α is open in X_α

and $\pi_\alpha : X \to X_\alpha$ is the Coordinate projection

Let U_α be open in $X_{\phi(\alpha)}$

Consider the subbasic open set in X_ϕ

$$V = \{y \in X_\phi : y_\alpha \in U_\alpha\}$$

$$f^{-1}(V) = \{x : X : f(x)_\alpha \in U_\alpha\}$$

$$= \{x \in X : x_\alpha \in U_\alpha\}$$

$$= \pi^{-1}{}_\alpha (U_\alpha)$$

Since U_α is open and π_α is continuous, the set is open in X

$\Rightarrow$ the inverse image of every subbasic open set is open

Consequently, f is continuous.

By the same argument, for every subbasic open set

$$W = \{x \in X : x_\alpha \in U_\alpha\}$$

we have $(f^{-1})^{-1}(W) = \{y \in X_\phi : y_{\phi^{-1}(\alpha)} \in U_\alpha\}$

this is a subbasic open set in $X\phi$

Hence f^{-1} is continuous.

We prooved that

f is bijective, Continuous and has a Continuous inverse

therefore $X \cong X\phi$

Question No. 4

Let • τ_1 denote product topology on Y

τ_2 denote the subspace topology on Y inherited from X

We shall proove $\tau_f = \tau_2$

Let B be a basic open set in X

then $B = \bigcap_{\alpha \in F} \pi^{-1}_\alpha (U_\alpha)$

where each U_α is open in X_α

By the definition of subspace topology, basic open set in Y is

$B \cap Y$

Now, $B \cap Y = Y \cap \bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha)$

$= \bigcap_{\alpha \in F} \left(Y \cap \pi^{-1}_\alpha (U_\alpha) \right)$

$$= \bigcap_{\alpha \in I} q_\alpha^{-1}(V_\alpha \cap Y_\alpha)$$

Since V_α is open in X_α the set $V_\alpha \cap Y_\alpha$ is open in Y_α by the subspace topology

Therefore, $B \cap Y$ is open in product topology on Y

Hence $\tau_2 \subseteq \tau_1$

Let C be a basic open set in product topology on Y

then, $C = \bigcap_{\alpha \in F} q_\alpha^{-1}(V_\alpha)$

where V_α is open in Y_α

Since Y_α has a subspace topology inherited from X_α,

$\exists$ an open set V_α in X_α such that

$$V_\alpha = U_\alpha \cap Y_\alpha$$

therefore,

$$C = \bigcap_{\alpha \in F} q_\alpha^{-1}(U_\alpha \cap Y_\alpha)$$

$$= Y \cap \bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha)$$

But $\bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha)$ is open in X

Because it is basic open set in the product topology on X

thus, Q is the intersection of Y with an open set in X

Hence Q is open in the subspace topology on Y

therefore

$$\tau_1 \subseteq \tau_2$$

so, $\tau_1 = \tau_2$

Question No: 5

By the definition of product topology, every basic open set has the form

$$B = \bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha)$$

where
- $F \subseteq A$ is finite
- Each U_α is open in X_α

Case 1: $\rightarrow \beta \in F$

then $B = \bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha)$

the image of B under p_β is

$$p_\beta(B) = U_\beta$$

provived the other coordinate restrictions are non empty

Indeed, every element of B has its β-coordinate in U_β

~~Conted~~ Conversely, any element of U_β can be extented to a point of B if all the other coordinate restrictions are non empty

Since U_β is open in X_β, the image is open

If another coordinate restriction is empty, then $B = \phi$ whose image is also open.

Case 2 :→ $\beta \notin \Gamma$

in this case, there is no restriction on the β-coordina

$$P_\beta(B) = X_\beta$$

provived $B \neq \phi$

Since X_β is open in itself, the Image is open

If $B = \phi$ its image is open again.

Now, $U = \bigcup_{\lambda \in \Lambda} B_\lambda$

Using the fact that images preserve arbitary unions we get

$$P_\beta(U) = P_\beta\left(\bigcup_{\lambda \in \Lambda} B_\lambda\right)$$

$$= \bigcup_{\lambda \in \Lambda} P_\beta(B_\lambda)$$

from above step, $P_\beta(B_\lambda)$ is open in X_β

An arbitary union of open sets is open

thus, $P_\beta(U)$ is open in X_β.

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part (ii) NO

take $X_1 = \mathbb{R}$, $X_2 = \mathbb{R}$ with their usual Euclidean topologies.

then $X = X_1 \times X_2 = \mathbb{R}^2$

Consider the coordinate projection $P_1 : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$p_1(x, y) = x$$

Consider

$$F = \{(x, y) \in \mathbb{R}^2 : xy = 1\}$$

Define $h : \mathbb{R}^2 \to \mathbb{R}$

by $h(x,y) = xy$

Since Multiplication is continuous, h is continuous

Also $F = h^{-1}(\{1\})$

Since $\{1\}$ is closed in $\mathbb{R}$ and the inverse image of closed set under a continuous function is closed.

We say F is closed in $\mathbb{R}^2$

we have $p_1(F) = \{x \in \mathbb{R} \mid \exists\, y \in \mathbb{R}$ such that $xy = 1\}$

$= \mathbb{R} \setminus \{0\}$

Indeed if $x \neq 0$ we can choose

$y = 1/x$

But if $x = 0$

No such y exists

therefore $p_1(F) = \mathbb{R} \setminus \{0\}$

the set $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$

is not closed in $\mathbb{R}$ because its complement

$\{0\}$ is not open in $\mathbb{R}$

thus, we found a closed set $F \subseteq \mathbb{R}^2$ whose

image under p_1 is not closed.

Consequently P_1 is not closed.

So, Every coordinate projection is an open Map

A coordinate projection need not be a closed Map.